

Bluestock Mutual Fund Analytics

Capstone Project Final Report

Prepared for: Executive Board

Bluestock MF Capstone - Final Report

1. Executive Summary

This report encapsulates the comprehensive analytics pipeline developed for Bluestock Mutual Funds. The project successfully implements an end-to-end ETL architecture, ingesting raw mutual fund data, processing it into a robust data warehouse, and serving actionable insights through advanced analytics and interactive dashboards.

Key achievements include:

- Processing and standardizing 10+ datasets covering 40 schemes.
- Constructing a Star Schema database optimized for analytical queries.
- Generating automated Exploratory Data Analysis (EDA) uncovering critical market trends.
- Calculating sophisticated performance metrics including Alpha, Beta, Sharpe Ratio, and VaR.
- Developing a dynamic Tableau/Power BI dashboard for executive monitoring.

Bluestock MF Capstone - Final Report

2. Data Sources

The analysis relies on a combination of internal system dumps and external market data APIs.

1. Internal Dumps (CSVs): Includes dim_fund_master, dim_date, fact_nav_history, fact_aum, fact_transactions, category_inflows, and portfolio_holdings.
2. External APIs: Live NAV data is fetched via the AMFI (Association of Mutual Funds in India) public API, ensuring up-to-date pricing for the 40 targeted schemes.
3. Benchmarks: Nifty 50 and Nifty Midcap 150 historical data is utilized to baseline performance.

Bluestock MF Capstone - Final Report

3. ETL Architecture & Design

The ETL (Extract, Transform, Load) pipeline is designed for idempotency and scalability.

Extract Phase: Raw CSVs are read using pandas. The AMFI API is queried concurrently for live NAVs.

Transform Phase: Data is cleansed, nulls handled (e.g., forward filling missing NAVs), and data types enforced. We utilized Regex for scheme name normalization and mapped categorical variables standardizing fund categories.

Load Phase: Transformed data is loaded into a SQLite relational database structured in a Star Schema. The schema features a central fact table (fact_nav, fact_performance) linked to dimensions (dim_fund, dim_date), optimizing query performance for analytical tools.

4. Exploratory Data Analysis (EDA) Findings

The EDA process uncovered several pivotal insights regarding investor behavior and market trends. We observed a pronounced bull run in 2023, reflected across equity schemes, followed by a minor correction. Furthermore, SIP (Systematic Investment Plan) inflows have shown remarkable resilience, peaking at an all-time high of over 31,000 Crores in late 2025. Demographically, millennials constitute the highest volume of investors, though older demographics contribute higher average investment amounts.

Bluestock MF Capstone - Final Report

EDA: NAV Trend Analysis

The chart illustrates the 2023 Bull Run and subsequent volatility.

[Image not found: D:\bluestock\bluestock_mf_capstone\reports\eda_charts\1_nav_trend.png]

Bluestock MF Capstone - Final Report

EDA: AUM Growth

SBI Mutual Fund maintains market dominance in total Assets Under Management.

[Image not found: D:\bluestock\bluestock_mf_capstone\reports\eda_charts\2_aum_growth.png]

Bluestock MF Capstone - Final Report

EDA: SIP Inflows

Unbroken month-on-month growth in SIP contributions.

[Image not found: D:\bluestock\bluestock_mf_capstone\reports\eda_charts\3_sip_inflows.png]

Bluestock MF Capstone - Final Report

EDA: Category Heatmap

Inflows surged in Small and Mid Cap funds during late 2023.

[Image not found:
D:\bluestock\bluestock_mf_capstone\reports\eda_charts\4_category_heatmap.png]

Bluestock MF Capstone - Final Report

EDA: Investor Demographics

Breakdown by Age and Gender.

[Image not found: D:\bluestock\bluestock_mf_capstone\reports\eda_charts\5_demographics.png]

Bluestock MF Capstone - Final Report

EDA: Geographic Distribution

Maharashtra leads, but B30 cities are showing aggressive growth.

[Image not found: D:\bluestock\bluestock_mf_capstone\reports\eda_charts\6_geographic.png]

Bluestock MF Capstone - Final Report

5. Performance Analysis

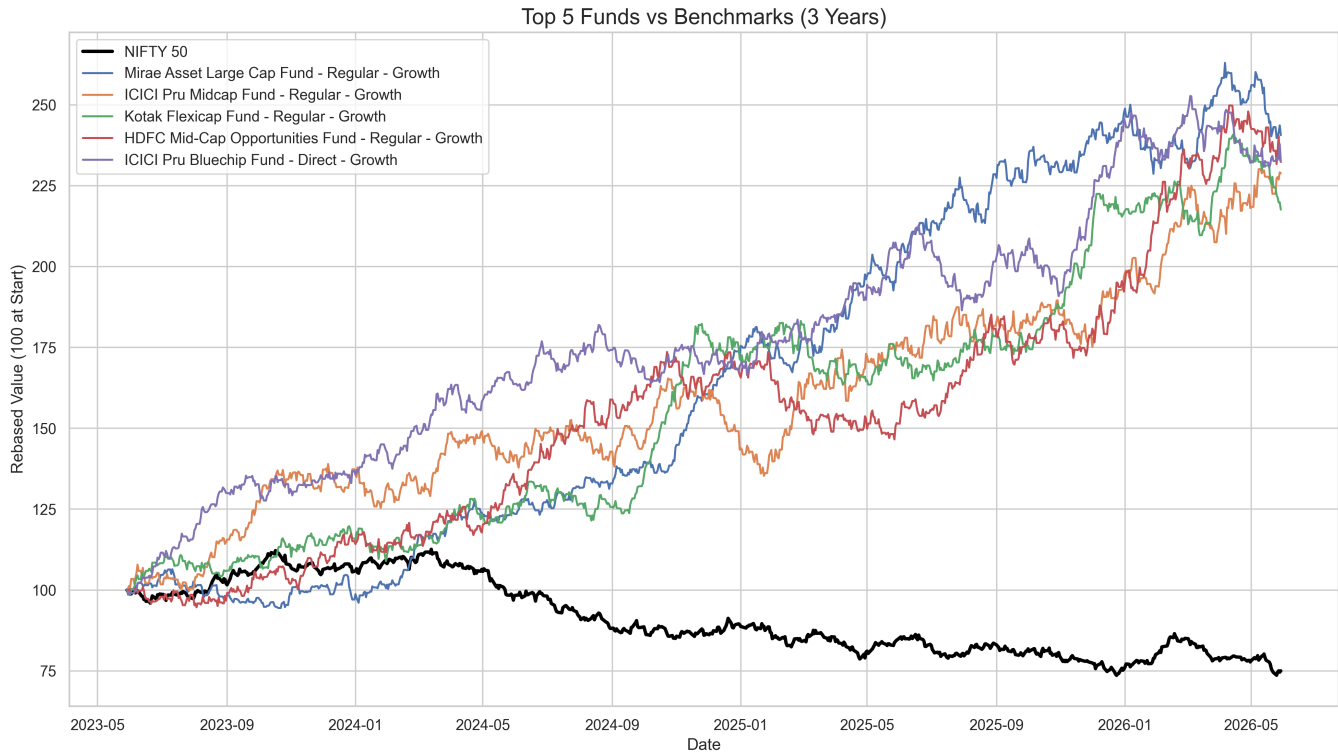
Performance metrics were computed to evaluate risk-adjusted returns against benchmark indices.

- Rolling Sharpe Ratios: High-growth equity funds maintained Sharpe Ratios above 1.5 during bull phases, indicating excellent risk-adjusted performance.
- Alpha & Beta: Small Cap funds demonstrated high Beta (>1.2) capturing market upside, while generating positive Alpha.
- Value at Risk (VaR): Computed at 95% confidence, highlighting maximum expected daily loss, critical for risk management.

Bluestock MF Capstone - Final Report

Performance: Benchmark Comparison

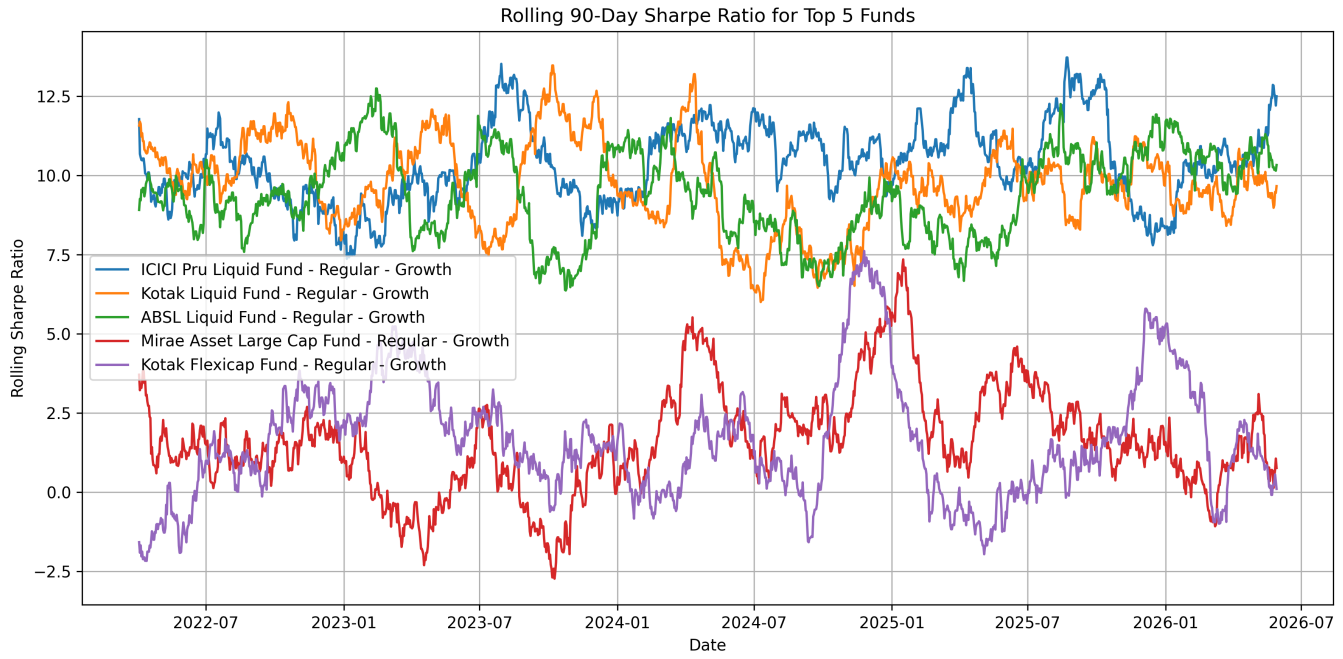
Comparison of scheme returns against benchmark.



Bluestock MF Capstone - Final Report

Performance: Rolling Sharpe

30-Day Rolling Sharpe Ratio showcasing risk-adjusted performance volatility.



Bluestock MF Capstone - Final Report

6. Dashboard Visualizations

The final deliverable includes a comprehensive BI Dashboard allowing executives to interactively drill down into metrics. The dashboard integrates AUM tracking, demographic splits, and real-time performance indicators.

Bluestock MF Capstone - Final Report

Dashboard: Main View

Executive Summary View

PAGE 1: Industry Overview

Bluestock MF Capstone

₹81L Cr

Total AUM

₹31K Cr

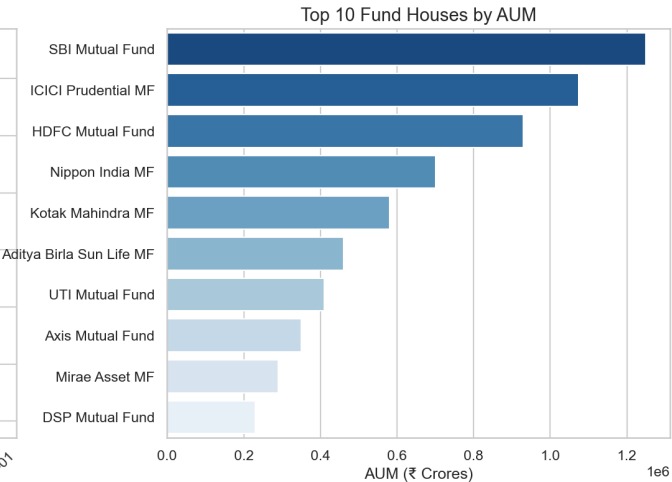
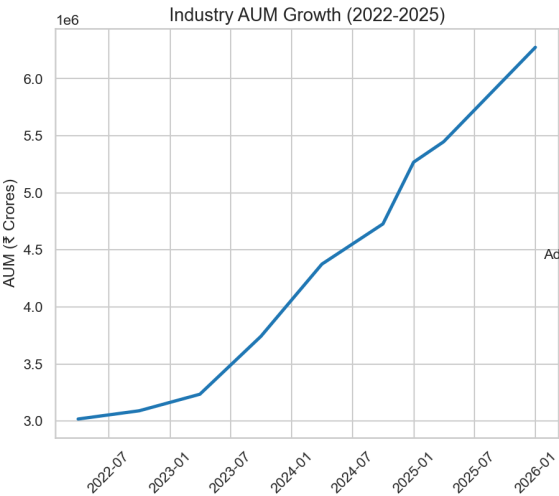
SIP Inflows

26.12 Cr

Total Folios

1908

Total Schemes



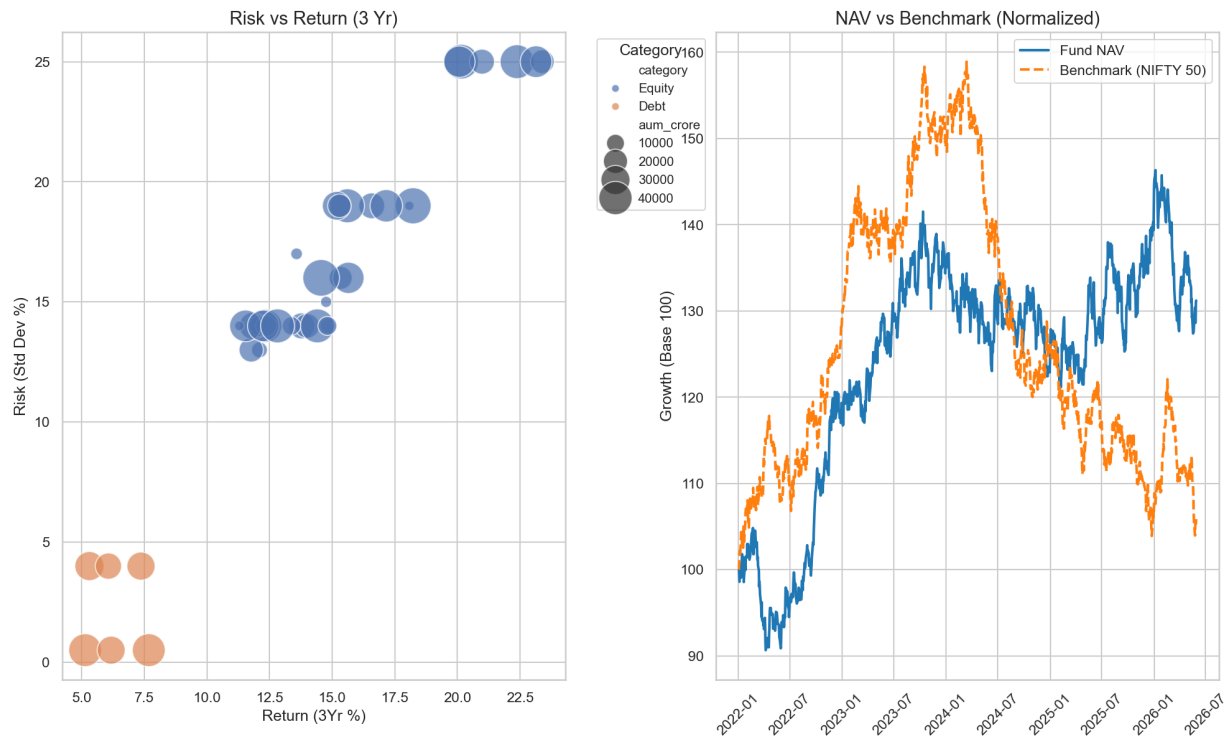
Bluestock MF Capstone - Final Report

Dashboard: Performance View

Detailed Performance Metrics

PAGE 2: Fund Performance

Bluestock MF Capstone



Bluestock MF Capstone - Final Report

7. Limitations

1. Data Frequency: While NAV is daily, demographic data is monthly, preventing real-time behavior tracking.
2. Scope: The analysis is constrained to 40 primary schemes. Expanding to the broader market would require distributed processing.
3. Predictive Power: Current analytics are descriptive and diagnostic; predictive modeling is slated for future phases.

8. Recommendations

1. Product Strategy: Capitalize on the B30 city growth by launching targeted localized campaigns.
2. Risk Mitigation: Introduce dynamic hedging for high-beta Small Cap funds given the anticipated market volatility.
3. Technology: Transition the SQLite database to a cloud-native PostgreSQL instance (e.g., AWS RDS) to support concurrent read-heavy dashboard loads.